

Divita Saraogi

Data Scientist | AI & Machine Learning Engineer

divitadsaraogi@gmail.com | +91-8847759169 | Bengaluru, India
LinkedIn: [linkedin.com/in/divita-saraogi](https://www.linkedin.com/in/divita-saraogi), Github: github.com/DivitaDSaraogi

PROFESSIONAL SUMMARY

Data Scientist with a PhD in Physics from IIT Bombay and 2+ years of industry experience developing AI, Machine Learning, Deep Learning, and statistical modeling solutions. Currently building spectral intelligence systems at Picarro for compound fingerprinting, speciation, and quantification using Python, PyTorch, and AWS. Experienced in large-scale data analytics, Bayesian statistics, scientific computing, and transforming complex datasets into scalable ML solutions.

TECHNICAL SKILLS

Programming & Data Analysis:	Python, SQL, R, Pandas, NumPy, Excel, C, C++
Machine Learning & AI:	PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Predictive Modeling, Feature Engineering
Data Visualization & BI:	Power BI, Matplotlib, Seaborn
Cloud, Tools & Development:	AWS, GitHub, Docker, PyCharm, Jupyter Notebook
Scientific & Research Computing:	Spectral Analysis, Scientific Data Processing, Statistical Modeling, Experimental Data Analysis
Statistics & Analytics:	Bayesian Statistics, Statistical Modeling, Hypothesis Testing, Experimental Design

PROFESSIONAL EXPERIENCE

Data Scientist – AI Team | Picarro Technologies India Pvt. Ltd.

Jun 2025 – Present | Bengaluru, India

- Developed AI-driven spectral analysis pipelines for compound speciation and quantification on datasets exceeding **100M+** spectral observations.
- Built ML and deep learning models for identification and classification of **600+** chemical compounds using high-dimensional spectroscopy data.
- Designed optimized instrument configuration pipelines for improved analytical performance and AI model validation.
- Developed scalable AWS-based workflows for data processing, feature engineering, and model experimentation.
- Collaborated with cross-functional teams to improve model accuracy, reliability, and deployment readiness.

Data Scientist | TossCall Services India Pvt. Ltd.

Feb 2024 – Feb 2025 | Bhillai, India

- Developed machine learning models for customer return prediction and business analytics.
- Built Power BI dashboards and automated reporting pipelines for KPI monitoring and customer behavior analysis.
- Performed EDA, feature engineering, and statistical analysis on large business datasets.
- Optimized SQL workflows and delivered actionable insights to business stakeholders.

PhD Research Scholar | Indian Institute of Technology Bombay

Jan 2018 – Jan 2024 | Mumbai, India

- Developed real-time Gamma-Ray Burst localization pipelines using AstroSat and Fermi satellite observations.
- Analyzed data from bright GRBs and developed automated pipelines for large-scale astrophysical datasets.
- Published peer-reviewed research in high-energy astrophysics and presented findings at international conferences.

EDUCATION

Ph.D. Physics, IIT Bombay (2024)

M.Sc. Physics, Sant Gadge Baba Amravati University (2014)

B.Sc. Physics, LAD College, Nagpur (2012)

PUBLICATIONS

- Authored multiple peer-reviewed publications in high-energy astrophysics, focusing on Gamma-Ray Burst localization, polarization, and statistical analysis.
- Presented research findings at international conferences and scientific workshops.

AWARDS & ACHIEVEMENTS

- ASI Best Poster Award (2024) for Gamma-Ray Burst localization research.
- Multiple peer-reviewed publications in high-energy astrophysics and statistical data analysis.

CERTIFICATIONS

Microsoft SQL | Python for Data Science | Power BI | Statistics & Mathematics for Data Science | Advanced Excel